Bálint Gyevnár

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Research Interests

Metascience; AI Scientists; Agentic Scientific Discovery; Explainable Multi-Agent Systems; Responsible AI; AI Safety; Human-AI Interaction.

Education

PhD in Natural Language Processing — Sep. 2021 – Jan. 2026
University of Edinburgh — *Edinburgh, UK*
Thesis: Action Explanation of Multi-Agent Systems via Counterfactual Reasoning
Supervisors: Stefano V. Albrecht, Shay B. Cohen, and Christopher G. Lucas

Integrated Master of Informatics — Sep. 2016 – May 2021
University of Edinburgh — *Edinburgh, UK*
Thesis: Comparison of Account Survival on Twitter During the First Wave of COVID-19
Supervisor: Maria Wolters

Exchange Year in Computer Science — Aug. 2018 – Jun. 2019
Nanyang Technological University — *Singapore*

Academic Experience

Postdoctoral Research Associate — Sep. 2025 – present
Carnegie Mellon University — *Pittsburgh, PA*

- Investigating **failure modes of agentic AI scientists**, such as data poisoning, reward hacking, spin doctoring, with actionable mitigations—work with *Nihar Shah* and *Atoosa Kasirzadeh*;
- Understanding the epistemic **landscape of AI hypothesis generation** and steering, e.g. using formalization of mathematics in Lean—work with *Simon DeDeo* and *David Barack*.

Teaching Assistant — Sep. 2020 – May 2025
University of Edinburgh — *Edinburgh, UK*

- Teaching assistant for courses in machine learning, computing systems, and two data analysis courses; Marker for natural language processing, reinforcement learning, machine learning courses.

Research Internship — May 2020 – Oct. 2020
Five AI Ltd. — *Edinburgh, UK*

- Development and evaluation of goal-based interpretable prediction and planning for autonomous vehicles;
- Main contributor of open-source implementation on GitHub with added support for CARLA.

Public Service

Academic Reviewing

- *Program committee member*: AAAI 2025, AIES 2025, FAccT 2026, XAI 2025;
- *Reviewer:* Nature Communications 2025, Artificial Intelligence Review 2025, International Journal of Human-Computer Interaction 2025, Transactions on Intelligent Transportation Systems 2025, CHI 2026, ECAI 2023, EMNLP2025, ICRA 2025, IASEAI 2026, IROS 2025, JAAMAS 2026, NeurIPS 2025, UKRI 2026, Survival and Flourshing Fund

Workshop Organiser Oct. 2025
Evaluating Explainable AI and Complex Decision-Making *ECAI 2025*

Teaching

- *Teaching Assistant (Spring 2024):* Evaluating Sustainable Lands and Cities; Data, Mobility, Infrastructure;
- *Marker (Spring 2022-24):* Machine Learning Theory, Reinforcement Learning, Natural Language Processing
- *Tutor (Autumn 2020):* Introductory Applied Machine Learning; Introduction to Computing Systems;

Awards

- Colours Award for Outstanding Volunteering Contribution to Sports;
  Edinburgh University Sports Union; Jun. 2024.
- Love, Sex, and AI — AI100 Early Career Essay Competition Featured Essay;
  One Hundred Year Study on Artificial Intelligence at Stanford University; Aug. 2023.
- £4,000 GBP Trustworthy Autonomous Systems Early Career Researcher Award;
  UK Research & Innovation; Jun. 2023.
- $1,000 USD Shape the Future of ITS Competition Award;
  IEEE Intelligent Transportation Systems Society; Aug. 2022.

Publications

Action Explanation of Multi-Agent Systems via Counterfactual Reasoning
B. Gyevnár
Doctoral Dissertation, University of Edinburgh, 2026

Distributed Denial of Science: How Indirect Data Poisoning of AI Systems Can Industrialize Scientific Fraud
B. Gyevnár, A. Kasirzadeh, N.B. Shah
arXiv preprint, 2026

Bridging the Gap in the Responsible AI Divides
B. Gyevnár, A. Kasirzadeh.
arXiv preprint, 2026

AI Epistemic Risks: Emerging Mechanisms & Evidence
M. Yang, S. Casper, J. Stray, J. Li, C. Jones, A. Gausen, N. Jaques, B. Christian, B. Gyevnár, H. Kirk, Z.He, D. Zhao, S.S. Looi, J. Levy, K. Hackenburg, E. Seger, M. Kowal, M. Malonza, L. Hewitt, H. Lin, M. Sap, D. Hadfield-Menell, T. Costello, R. Rabbany, J.-F. Godbout, D. Rand, A. Kasirzadeh, G. Pennycook, Y. Bengio, K. Pelrine
SSRN preprint, 2026

Integrating Counterfactual Simulations with Language Models for Explaining Multi-Agent Behaviour
B. Gyevnár, C.G. Lucas, S.V. Albrecht, S.B. Cohen.
AAMAS 2026

AI Safety for Everyone
B. Gyevnár, A. Kasirzadeh.
Nature Machine Intelligence, 2025

Objective Metrics for Human-Subjects Evaluation in Explainable Reinforcement Learning
B. Gyevnár, M. Towers.
RLDM 2025

People Attribute Purpose to Autonomous Vehicles When Explaining Their Behavior: Insights From Cognitive Science for Explainable AI
B. Gyevnár, S. Droop, T. Quillien, S.B. Cohen, N.R. Bramley, C.G. Lucas, S.V. Albrecht.
CHI 2025

Causal Explanations for Sequential Decision-Making in Multi-Agent Systems
B. Gyevnár, C. Wang, S.B. Cohen, C.G. Lucas, S.V. Albrecht.
AAMAS 2024

Explainable AI for Safe and Trustworthy Autonomous Driving: A Systematic Review
A. Kuznietsov*, B. Gyevnár*, C. Wang, S. Peters, S.V. Albrecht. [* equal contribution]
IEEE T-ITS, 2024

Bridging the Transparency Gap: What Can Explainable AI Learn From the AI Act?
B. Gyevnár, N. Ferguson, B. Schafer.
ECAI 2023

Deep Reinforcement Learning for Multi-Agent Interaction
I.H. Ahmed, C. Brewitt, I. Carlucho, F. Christianos, M. Dunion, E. Fosong, S. Garcin, S. Guo, B. Gyevnár, T. McInroe, G. Papoudakis, A. Rahman, L. Schäfer, M. Tamborski, G. Vecchio, C. Wang, S.V. Albrecht.
AI Communications, 2022

Communicative Efficiency or Iconic Learning: Do Communicative and Acquisition Pressures Interact to Shape Colour-Naming Systems?
B. Gyevnár, G. Dagan, C. Haley, S. Guo, F. Mollica.
Entropy, 2022

A Human-Centric Method for Generating Causal Explanations in Natural Language for Autonomous Vehicle Motion Planning [best paper runner-up]
B. Gyevnár, M. Tamborski, C. Wang, C.G. Lucas, S.B. Cohen, S.V. Albrecht.
IJCAI 2022 Workshop on Artificial Intelligence for Autonomous Driving

Interpretable Goal-based Prediction and Planning for Autonomous Driving S.V. Albrecht, C. Brewitt, J. Wilhelm, B. Gyevnár, F. Eiras, M. Dobre, S. Ramamoorthy.
ICRA 2021

GRIT: Fast, Interpretable, and Verifiable Goal Recognition with Learned Decision Trees for Autonomous Driving
C. Brewitt, B. Gyevnár, S. Garcin., S.V. Albrecht.
IROS 2021

Invited Talks

AI in scientific research: what could go wrong and how to fix it?
Machine Learning Summer School, Columbia Univeristy, 2026.

Human and AI solution paths in formalizing expert mathematics
Social Reasoning and the Ecology of Thought, IVADO & MILA, 2026.

AI safety for everyone
Ninth Workshop of the Center for Human-Compatible AI, UC Berkley, 2025.

How do we make explainable AI work for people?
Talk at the Machine Learning and Modelling Seminar, Charles University of Prague, 2024.